5-11-24

Unit - 2

Relations

Cartesian Product.

$A = \{1, 2, 3\}$, $B = \{x, y, z\}$

$A \times B = \{(1,x), (2,x), (3,x), (1,y), (2,y), (3,y), (1,z), (2,z)(3,z)\}$.

The cartesian product of two sets A & B denoted as $(A \times B)$ is the set of all the ordered pairs (a, b) where "a" element of set A and "b" is an element of set B.

Note $A \times B = \{(a,b) : a \in A \text{ and } b \in B\}$

Note
- Each element in $A \times B$ is ordered pair where the orders matter. The pair (a, b) is distinct from (b, a) unless $a = b$.
- The result of $A \times B$ is itself a set containing all possible combination of $A \times B$.
- If either set A or set B is empty $A \times B$ is also empty.

Relations

- A relation is a way to establish a connection b/w Elements of two sets.
- A relation from Set A and Set B is a subset of the cartesian product.
- Relation are always represented with 'R', $R \subseteq A \times B$,
- If $(a, b) \in R$, we say a is related to b and is represented as aRb.

Example :- $R = \{(a, a) : a \in A\}$

set $A = \{1, 2, 3\}$. $R = \{(1,1), (2,2), (3,3)\}$

Q Establish the elements for a ~~elements~~ relation, $R = \{(a, b) : a \text{ divides } B\}$ on the set $A = \{1, 2, 3\}$ & $B = \{4, 5, 6\}$

$R = \{(1,4), (1,5), (1,6), (2,4), (2,6), (3,6)\}$.

Properties of Relations

1) **Reflexive Relation** :- A relation R on a set A is called reflexive, if

$\forall x \in A$ [for all x belongs to A]

$(x, x) \in R$

Ex Let A{ 1, 2, 3, 4, 5}

$R = \{(1,1), (2,2), (3,3), (4,4)\}$ ✓ Reflexive

$R = \{(1,1), (3,3), (4,4)\}$ ✗ Not Reflexive.

↳ should contain (x,x) ~~for~~ of every element of A

2) **Symmetric Relation** :- A Relation R on set A is called symmetric if $(x, y) \in R$, (y, x) should also belongs to R. for all the value of x & y belong to A.

Q set $A = \{1, 2, 3, 4\}$ Generate the relation.

1. It is symmetric not reflexie.
2. Reflexive by not symmetric
3. Both Reflexive and Symmeteric
4. Neither Reflexive or Symmetric.

soln

1. $\{(1,2), (2,1)\}$
2. $\{(1,1), (2,2), (3,3), (4,4), (1,2)\}$.
3. $\{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1)\}$
4. $\{(1,2), (1,3), (1,4)\}$.

3) Transitive property :- A relation R on set A is called transitive if $\forall (x, y, z) \in A$,

if (x, y) & $(y, z) \in R$.

Then, (x, z) should belong to R.

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Anti - Symmetry :- Given a Relation R on set A, R is called anti symmetric, if for all A, B belong A ($\forall A, B \in A$) i.e. A is related to (ARB), & B is related to A (B to A) implies

$A = B$.

Q $A\{1, 2, 3\}$, $R = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

$(1,2)$, $(2,1)$
$(1,3)$, $(3,1)$
$(2,3)$, $(3,2)$

Not Anti symmetric
Because, $(1 \neq 2)$
$2 \neq 3$
$1 \neq 3$

Q Which of the following are anti symmetric.

$R_1 = \{(1,1), (1,3), (3,1)\}$ | $R_2 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$

$R_3 = \{(1,1), (2,2), (3,3), (4,4)\}$. | $R_4 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,4), (4,1)\}$.

$R_1 \Rightarrow$ Symmetric, ~~A~~, $R_2 \rightarrow$

Partial Order Relations

If Relation "R" is defined on set "A", Relation R will be called partial order relation, If R is reflexive, Antisymmetric and transitive.

Q Given set $A = \{1,2,3,4\}$, $R_1 = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)\}$

$R_2 \Rightarrow \{(1,1),(1,2),(2,1)\}$.

$R_3 \Rightarrow \{(1,1),(1,2),(1,4),(2,1),(2,2),(3,3),(4,1),(4,4)\}$

$R_4 \Rightarrow \{(2,1),(3,1),(3,2),(4,1),(4,2),(4,3)\}$
$(2,2)$,

$R5 \Rightarrow \{(1,1)(1,2),(1,3)(1,4)(2,2)(2,3)(2,4)(3,3)$
$(3,4),(4,4)\}$

determine which of the property of the relation are satisfied by any relation, is there any relation which is POR.

R_1	R_2	R_3	R_4	R_5
$\Rightarrow$ Reflexive ✗	$\Rightarrow$ Sym ✓	$\Rightarrow$ Reflexive ✓	$\Rightarrow$ Ref ✗	Trans
$\Rightarrow$ symmetry ✓	trans ✓	Transitive ✓	Trans ✓	Reflexive.
$\Rightarrow$ transitive ✓	Anti Sym ✗	symmetric ✓		
$\Rightarrow$ Anti Symmetry ✗	Reflexive ✗	AntiSym ✗		

- $R_4 \Rightarrow \{(4,2),(2,1),(4,1)\} \rightarrow$

  $\{(4,3),(3,1),(4,1)\}$

  $\{(3,2),(2,1),(3,1)\}$

- for Anti symmetry, to happen relation should not be like ..., Like, R contains (1,2) but (2,1), so ... the symm, but it contains (1,1) so will it be Anti Symm.

(To show anti symmetry in Relation just show Reflexive bcz Antisymmetry is Reflexive only)

(& To show transitive give only one example)

Ex of Partial order. $\{(1,1),(2,2),(3,3),(1,2),(2,3),(1,3)\}$
$A = (1,2,3)$

* Equivalence relation: A Relation R on set A is said to be equivalence, if it satisfy the relation property of Reflexive, Symmetry, transitive

q $A = \{1,2,3\}$. $R_1 \{(1,1), (2,2), (2,3), (3,2), (3,3)\}$

$R_2 = \{(1,1), (1,2), (1,3), (2,2), (3,3)\}$ ~~(2,1), (2,3)~~

$R_3 \Rightarrow \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,2), (3,3)\}$ (3,1)

* $R_1 = \{(1,1), (2,2), (2,3), (3,2), (3,3)\}$.

- $\forall a \in R$, $(a,a) \in R$.

  $\forall 1,2,3$ $(1,1), (2,2), (3,3) \in R$,

  $\therefore$ Relation R_1 is Reflexive.

- $\forall a,b \in R$, $(a,b), (b,a) \in R$.

  $\forall 1,2,3$, $(2,3), (3,2) \in R$.

- $\forall a,b,c \in R$, $(a,b), (b,c) \in R$, it must include $(a,c) \in R$.

  $\forall (1,2,3) \in R$, $(2,3), (3,2) \in R$, $(2,2) \in R$.

so, Relation R_1 is Reflexive, symmetric, transitive. i.e. R_1 is equivalent.

* $R_2 = \{(1,1), (1,2), (1,3), (2,2), (3,3)\}$

- $\forall a \in R$, $(a,a) \in R$.

  $\forall (1,2,3) \in R$, $\{(1,1), (2,2), (3,3)\} \in R$. $\therefore$ Relation R_2 is Reflexive.

- $\forall (a,b) \in R$, $(a,b) \in R$, $(b,a) \in R$.

  $\forall (1,2,3) \in R$, $\{$

- Q given an ex of Relation R on A such that R is reflexive & symmeteric

A = {1, 2, 3}

R is irreflexive, symm, not transitive

R is sym, trans, not reflexive.

R is Ref, symm, Not trans.

R is ~~irr~~ Reff, trans, not symm.

(1, 1) (1, 2) (1, 3)
(2, 1) (2, 2) (2, 3)
(3, 1) (3, 2), (3, 3).

$R_1 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$

$R_2 = \{(1,2), (2,1)\}$

$R_3 = \{(1,2), (2,1), (1,1)\}$

$R_4 \Rightarrow \{(1,1), (2,2), (3,3), \}$ ~~(1,2)~~ (1, 2), (2, 3), (2, 1), (3, 2)}

$R_5 \Rightarrow \{(1,2), (2,3), (1,3)\}$

15-11-024

Composite of Relation

If a, b, c are the sets with R_1 being a subset of $A \times B$ & R_2 being a subset of $B \times C$. then the composite relation $R_1 O R_2$, is a relation from set $A \rightarrow$ set C defined by $R_1 O R_2$ in such a way $R_1 O R_2 = \{ (x, z) \mid x \in A, z \in C$
$\exists (y \in B)$ with $(x, y) \in R$ and
$(y, z) \in R_2 \}$

example set $A = \{1, 2, 3, 4\}$, set $B \Rightarrow \{w, x, y, z\}$, set $C = \{5, 6, 7\}$.
consider $R_1 = \{(1, x), (2, x), (3, y), (3, z)\}$; $R_2 = \{(w, 5), (x, 6)\}$

$R_1 O R_2 = \{(1, 6), (2, 6)\}$

$R_3 = \{(w, 5) \,\&\, (w, 6)\}$

find $R_1 O R_3 = \{\}$ null set

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set $A = \{1, 2, 3, 4\}$ relation $R = \{(1, 2), (1, 3), (2, 4), (1, 4)\}$
relation $S = \{(1, 1), (1, 2), (1, 3), (2, 3), (2, 4)\}$, find the composite RoS & SoR

$ROS \Rightarrow \{(1, 3), (1, 4)\}$, ~~ROS~~ $SOR = \{(1, 1), (1, 4), (2, 4)\}$

Power's of Relation.

Given a set A and a relation R on A we define the powers of relation R recursively by

① $R^1 = R$

② For $n \in Z^+$, $R^{n+1} = ROR^n$

Q set $A = \{0, 1, 2, 3\}$, set $B = \{0, 2, 3\}$, $R = \{(1, 1), (2, 1), (3, 2), (2, 3)\}$

find the power of R^n where $n = 2, 3, 4, 5$

$R^{n+1} = ROR^n$

$R^2 = ROR$

$R^3 = ROR^2$

$R^4 = ROR^3$

$R^5 = ROR^4$

$R^2 \Rightarrow R \circ R \Rightarrow \{(1,1),(2,1),(3,1),(3,3),(2,2)\}$

$R^3 \Rightarrow R \circ R^2 \Rightarrow \{(1,1),(2,1),(3,1),(3,2),(2,3)\}$, $\Rightarrow \{(1,1),(2,1),(3,2),(2,3)\}$

$R^4 \Rightarrow R \circ R^3 \Rightarrow \{(1,1),(2,1),(3,1),(3,3),(2,2)\}$

$R^5 = R \circ R^4 = \{(1,1),(2,1),(3,1),(3,2),(2,3)\}$.

Q What is composite of $R_2 \circ R_1$ if R_1 is relation from set $\{1,2,3\}$ to set $\{1,2,3,4\}$ with $R_1 = \{(1,1),(1,3),(1,4),(2,3),(3,1),(3,4)\}$

& R_2 is a relation from set $R_2\{1,2,3,4\}$ $R_2 = \{(1,0),(3,1),(2,0),(3,2),(4,1)\}$.

$R_2 \circ R_1 \Rightarrow \{(3,1),(3,3),(3,4),(3,3),(4,1),(4,3),(4,4)\}$.

Q let $A = \{1,2,3\}$ set $B\{w,x,y,z\}$, set $C = \{4,5,6\}$ define the relation R_1 which subset of $A \times B$, R_2 subset of $B \times C$, ~~where the~~ R_3 subset $B \times C$

where $R_1\{(1,w),(2,w),(3,x),(1,y)\}$ & $R_2 = \{(w,5),(x,6),(y,4),(y,6)\}$

$R_3 = \{(w,4),(w,5),(y,6)\}$ determine

1. $R_1 \circ (R_2 \cup R_3)$ and $(R_1 \circ R_2) \cup (R_1 \circ R_3)$

2. $R_1 \circ (R_2 \cap R_3)$ and $(R_1 \circ R_3) \cap (R_1 \circ R_3)$

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1. $R_1 \circ (R_2 \cup R_3)$ and $(R_1 \circ R_2) \cup (R_1 \circ R_3)$.

$R_2 \cup R_3 \Rightarrow \{(w,5),(x,6),(y,4),(y,6),(w,4),(w,5),(y,6)\}$

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0–1 Matrix.

An $m \times N$ (0 – 1 matrix) $E = (e_{ij})_{m \times n}$ is a rectangular Array of no. arranged in m rows and n colums. where each e_{ij} for $[1 \le i \le m$ & $1 \le j \le m]$ denotes the entry in i^{th} row and j^{th} column of E. And each such entry is 0 or 1

m – rows → i, n → columns → j.

$$\begin{array}{c} \\ w \\ x \\ y \end{array}\begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$$

$R = \{ (w,1), (x,2), (x,3), (y,1) \}$.

Q Let set $A = \{1,2,3,4\}$, Set $B = \{w, x, y, z\}$, set $C = \{5,6,7\}$, $R_1 = \{(1,x), (2,x), (3,y), (3,z)\}$, $R_2 = \{(w,5), (x,6)\}$, $R_3 = \{(w,5), (w,6)\}$.

R_1 is subset of $A \times B$ ($R_1 \subseteq A \times B$), R_2 is subset of $B \times C$ ($R_2 \subseteq B \times C$), R_3 is subset of $B \times C$ ($R_3 \subseteq B \times C$)

find

① $M(R_1)$

② $M(R_2)$

③ $M(R_3)$

④ $M(R_1 \times R_2)$ Multiplication

⑤ $M(R_1 \text{ OR } R_2)$

consider M to be (0, 1) matrix.

$$M(R_1) \quad \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\begin{array}{c} \begin{array}{cccc} w & x & y & z \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{m_1 \times n_1} \end{array}$$

$$M(R_2) = \begin{array}{c} \\ 5 \\ 6 \\ 7 \end{array}\begin{array}{c} \begin{array}{cccc} w & x & y & z \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{m_2 \times n_2} \end{array}$$

$$MR_3 = \begin{array}{c} \\ 5 \\ 6 \\ 7 \end{array}\begin{array}{c} \begin{array}{cccc} w & x & y & z \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{m_3 \times n_3} \end{array}$$

(4) $M(R_1 * R_2) = \begin{array}{c|ccc} & 5 & 6 & 7 \\ \hline 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 3 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{array}$ (5)

(5) $R_1 O R_2 = \{(1,6),(2,6)\}$

$M(R_1 O R_2) = \begin{array}{c|ccc} & 5 & 6 & 7 \\ \hline 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 3 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{array}$

Hence ~~M of R~~ $M(R_1 \times R_2) = M(R_1 O R_2)$.

Q Let $A = \{1,2,3,4\}$, $R = \{(1,2),(1,3),(2,4),(3,2)\}$. where R is a matrix relation of 4×4 (0,1) matrix. where the entries are $m(i o j)$ with i & j in the range of 1 to 4.

(i) $M(R)^2$ (ii) $M(ROR)$

$A = \{1,2,3,4\}$. $R = \{(1,2),(1,3),(2,4),(3,2)\}$.

$M(R) \Rightarrow \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 0 & 1 \\ 3 & 0 & 1 & 0 & 0 \\ 4 & 0 & 0 & 0 & 0 \end{array}$ $\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

~~M(R o R)~~ $\Rightarrow$
$M(R) \times M(R)$ $\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 1 \\ 4 & 0 & 0 & 0 & 0 \end{array}$

$M(R O R) = M(R) \times M(R)$